

Supplementary Material:

Physics-Informed Wi-Fi-Based Device-Free Human Activity Recognition

(IEEE Sensors Journal Survey)

Zhihao Zhao et al.

This document accompanies the IEEE Sensors Journal review manuscript *Physics-Informed Wi-Fi-Based Device-Free Human Activity Recognition: A Survey*. It contains technical notes, case-by-case formulations, pseudocode, equations, and figures relocated from the main text to meet the review-paper page limit. The same content is versioned at https://github.com/zhihaozhao/Pi-Fi_Survey.

Contents

1	Relocated Main-Text Figures and Algorithms	3
1.1	Composite Maxwell-PINN objective (main-text equation)	3
2	Technical Notes for Reproducibility and Interpretation	3
2.1	Maxwell-PINN Formulations Used in Prior Work	3
2.1.1	Original PINN Framework [5]	3
2.1.2	Mapping PINN Formulations to Wi-Fi CSI: An Illustrative Survey Interpretation	4
2.2	Information-Theoretic Formulations for Wi-Fi sensing	7
2.2.1	Shannon’s Classical Information Theory	7
2.2.2	Mapping Shannon Theory to Wi-Fi sensing: Survey Interpretation of Cited Formulations	7
2.3	Cross-Domain Adaptation with Physical Invariance	11
2.3.1	Classical Domain Adaptation Theory	11
2.3.2	How Reviewed Studies Apply Physical Invariance to Domain Adaptation	11
3	Case-by-Case Formulations from Representative Studies	13
3.1	Case 1: Cross-Domain Physical Invariance Formulation	13
3.1.1	Original AirFi Implementation [17]	13
3.1.2	Formulation Notes	14
3.2	Case 2: Compression and Recognition Trade-off Formulation	14
3.2.1	Original EfficientFi Implementation [9]	14
3.2.2	Formulation Notes	15
3.3	Case 3: Phase Reconstruction Formulation	16
3.3.1	Original WiPhase Implementation [20]	16
3.3.2	Formulation Notes	16

3.4	Case 4: Sparse Geometric Modeling Formulation	17
3.4.1	Original WiHGR Implementation [22]	17
3.4.2	Formulation Notes	18
3.5	Case 5: Feature Decoupling Formulation	18
3.5.1	Formulation Notes	18
3.6	Case 6: Physics-Constrained Learning Formulation	19
3.6.1	Composite Physics-Constrained Formulation	19
4	Integration Summary	20
4.1	Cross-Reference Map of Appendix Notes and Cases	20
5	Appendix Algorithms	20
5.1	Adaptive Residual Sampler (ARS)	20

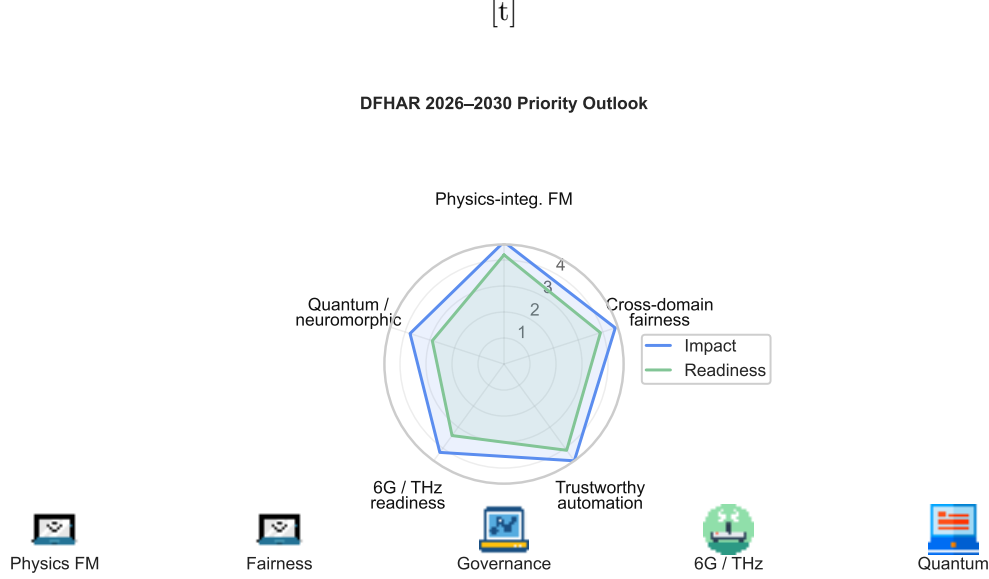


Figure 1: Impact versus readiness outlook distilled from recent DFHAR field reports [1–4]. Scores summarize reported benefit and deployment maturity in those studies; the chart is an evidence summary for open problems, not a prescriptive roadmap or cross-study ranking.

1 Relocated Main-Text Figures and Algorithms

Algorithm 1 Illustrative Maxwell-informed training loop (condensed from cited formulations)

Require: CSI $\{\mathbf{H}_{raw}\}$, material parameters $\{\epsilon_r, \mu_r\}$, loss weights $\{\lambda_i\}$

- 1: Sample collocation points $(\mathbf{r}, \omega)$; evaluate residual vector $[f_1, f_2, f_3, f_4]^\top$ (Eq. (1) and Supplementary Material)
 - 2: Minimise composite loss; log telemetry $\{\mathcal{L}_{physics,i}\}$; pause if residual exceeds envelope
 - 3: Export physics-consistent representations $\mathbf{z}_{physics}$ to downstream adaptation and compression pipelines
-

1.1 Composite Maxwell-PINN objective (main-text equation)

$$\mathcal{L}_{PINN} = \mathcal{L}_{data} + \sum_{i=1}^5 \lambda_i \mathcal{L}_{physics,i}, \quad (1)$$

2 Technical Notes for Reproducibility and Interpretation

This appendix provides technical notes that support interpretation of the survey synthesis in the main text, clarifying notation, assumptions, and formulations drawn from the cited prior work.

2.1 Maxwell-PINN Formulations Used in Prior Work

2.1.1 Original PINN Framework [5]

The PINN framework formulated by Raissi et al. [5] incorporates physical laws into neural network training via residual losses:

	Methods	Systems	Diagnostics
Ext. 1 Phys.-invariant	Residual gating; Ben-David (App. A-C)	AirFi, WiSR, WiCAU, MetaFormer	Residual telemetry; zero- shot KPIs
Ext. 2 Compression	MI guard; joint compression (App. A-B)	EfficientFi, CAPC, MaskCAE	MI telemetry; compression KPIs
Ext. 3 Phase-coherent	Phase-ratio + DTW (App. B-C)	WWiPhase, SHARP	Latency; coherence traces
Ext. 4 Sparse geometry	Path pursuit; SV sparsity (App. B-D)	WiHGR, Wilmg2.0	Packet refresh logs
Ext. 5 Decoupling	Orthogonal replay (App. B-E)	CUDSG, AutoDLAR	Leakage; fairness monitors
Ext. 6 Phys.-constrained	Residual sampling; PINNs (App. A-A)	TMANN, ARS, PINNsFormer	Sampler logs; edge schedule

[!t]

Figure 2: Conceptual swimlane map linking appendices, six survey portfolios, representative systems, and field diagnostics (telemetry, calibration, governance). Complements Table ??; illustrative layout, not measured runtime data.

PINN residual loss formulation from Raissi et al. [5]:

$$u_t + \mathcal{N}[u] = 0, \quad x \in \Omega, \quad t \in [0, T] \quad (2)$$

$$f := u_t + \mathcal{N}[u] \quad (3)$$

$$MSE = MSE_u + MSE_f \quad (4)$$

where:

$$MSE_u = \frac{1}{N_u} \sum_{i=1}^{N_u} |u(t_i^u, x_i^u) - u_i|^2$$

$$MSE_f = \frac{1}{N_f} \sum_{i=1}^{N_f} |f(t_i^f, x_i^f)|^2$$

2.1.2 Mapping PINN Formulations to Wi-Fi CSI: An Illustrative Survey Interpretation

Note: The steps below illustrate how studies in the reviewed corpus apply the Raissi et al. [5] framework to Wi-Fi CSI. The notation and parameter mappings are adopted from those cited sources, and the steps present an analytical reading of how those designs are structured within that framework.

Illustrative mapping from cited formulations:

Step 1: Identify Wi-Fi CSI as electromagnetic field quantities For Wi-Fi sensing, Channel State Information represents the complex electromagnetic field response [6]:

$$\begin{aligned} H(f, t) &= |H(f, t)| e^{j\angle H(f, t)} \\ &= \mathbf{E}_{\text{complex}}(\mathbf{r}, \omega, t) \cdot \mathbf{H}_{\text{complex}}(\mathbf{r}, \omega, t) \end{aligned}$$

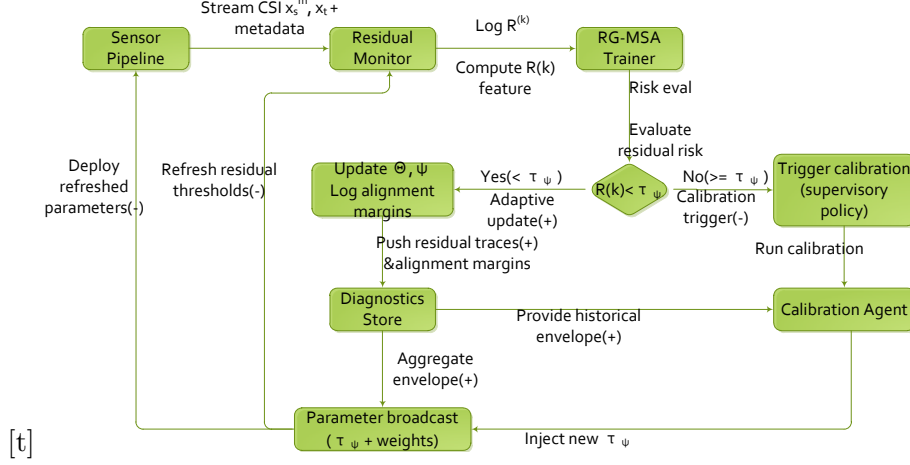


Figure 3: Illustrative sequence for residual-gated multi-source adaptation (sensor pipeline, residual monitor, trainer, calibration agent, diagnostics store), condensed from cited formulations. Not measured deployment telemetry.

Step 2: Apply Maxwell equations as physical constraints Maxwell equations in frequency domain (universal electromagnetic laws) [6]:

$$\nabla \times \mathbf{E} + j\omega\mu\mathbf{H} = 0 \quad (\text{Faraday's law}) \quad (5)$$

$$\nabla \times \mathbf{H} - j\omega\varepsilon\mathbf{E} = \mathbf{J} \quad (\text{Ampère's law}) \quad (6)$$

$$\nabla \cdot (\varepsilon\mathbf{E}) = \rho \quad (\text{Gauss's law}) \quad (7)$$

$$\nabla \cdot (\mu\mathbf{H}) = 0 \quad (\text{No magnetic monopoles}) \quad (8)$$

Step 3: Wi-Fi-sensing studies extend Raissi's residual approach to Maxwell constraints Studies such as Olivares et al. [6] and Shi et al. [7] apply Raissi's residual construction to electromagnetic field quantities, defining constraint functions analogously to $f := u_t + \mathcal{N}[u]$:

$$f_1 := \nabla \times \mathbf{E} + j\omega\mu\mathbf{H}$$

$$f_2 := \nabla \times \mathbf{H} - j\omega\varepsilon\mathbf{E} - \mathbf{J}$$

$$f_3 := \nabla \cdot (\varepsilon\mathbf{E}) - \rho$$

$$f_4 := \nabla \cdot (\mu\mathbf{H})$$

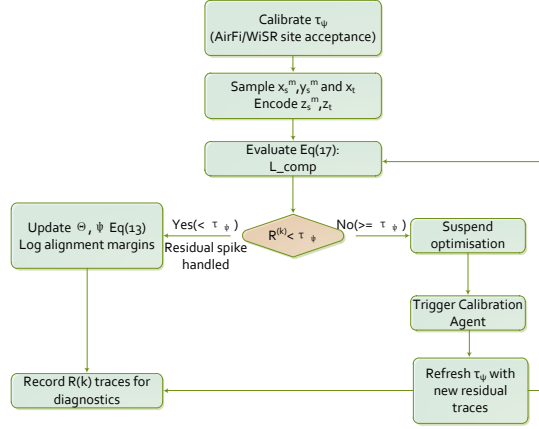
Step 4: Complete Maxwell-PINN loss function as used in reviewed studies Following the pattern of Eq. 4, Wi-Fi-sensing implementations [6,7] compose the electromagnetic constraints into a total loss:

$$\mathcal{L}_{Maxwell-PINN} = \mathcal{L}_{data} + \sum_{i=1}^4 \lambda_i \mathcal{L}_{physics,i}$$

$$\mathcal{L}_{physics,i} = \frac{1}{N_i} \sum_{j=1}^{N_i} |f_i(\mathbf{r}_j, \omega_j)|^2$$

Variable acquisition and implementation strategy:

- $\mathcal{L}_{data}$: Standard supervised learning loss (cross-entropy for activity classification), computed from labeled Wi-Fi CSI data and activity ground truth



[t]

Figure 4: Illustrative control flow for residual-gated adaptation: calibration, residual checks, and pause–recalibrate when drift exceeds the site-calibrated envelope (after AirFi/WiSR-style reports).

- $\lambda_i \in [10^{-4}, 10^{-1}]$: Physics constraint weighting parameters, typically set empirically through grid search or learned adaptively during training
- N_i : Number of spatial-frequency sampling points for the i -th Maxwell equation, typically 1000-5000 points uniformly sampled in the Wi-Fi coverage area
- $\mathbf{r}_j \in \mathbb{R}^3$: Spatial sampling points in meters, covering the room/environment geometry where Wi-Fi sensing occurs (e.g., $3\text{m} \times 3\text{m} \times 3\text{m}$ room)
- ω_j : Frequency sampling points covering Wi-Fi bandwidth, typically 20-160 MHz around carrier frequency
- $f_i(\mathbf{r}_j, \omega_j)$: Maxwell constraint violation at sampling point (j), computed using neural network predictions of electromagnetic fields

Practical implementation approach:

1. Field approximation: Neural network predicts simplified electromagnetic field components from CSI measurements
2. Constraint evaluation: Maxwell equations evaluated at discrete sampling points rather than continuous space
3. Parameter estimation: Material properties $\mu(\mathbf{r}), \epsilon(\mathbf{r})$ either preset with typical indoor values or learned during training
4. Adaptive weighting: Physics loss weights λ_i adjusted based on constraint violation magnitude to maintain numerical stability

In this illustrative mapping, each residual term $f_i = 0$ represents the Maxwell constraint as formulated in the cited studies [6, 7]; practical CSI implementations enforce approximations of these terms under simplifying assumptions about indoor propagation.

2.2 Information-Theoretic Formulations for Wi-Fi sensing

The compression-centric studies reviewed in Section ?? invoke Shannon’s mutual-information framework to motivate why selective quantization can preserve recognition accuracy; the key quantities are recalled here.

2.2.1 Shannon’s Classical Information Theory

From Cover & Thomas [8], the fundamental information-theoretic quantities:

Mutual Information (Cover & Thomas, 1999, Chapter 2):

$$I(X; Y) = \iint p(x, y) \log \frac{p(x, y)}{p(x)p(y)} dx dy \quad (9)$$

$$I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X) \quad (10)$$

Channel Capacity (Cover & Thomas, 1999, Chapter 7):

$$C = \max_{p(X)} I(X; Y) \quad (11)$$

2.2.2 Mapping Shannon Theory to Wi-Fi sensing: Survey Interpretation of Cited Formulations

Note: The variable mapping below follows how compression and sensing studies in the reviewed corpus (e.g., EfficientFi [9], Chen et al. [10]) instantiate Shannon’s framework. This reading draws on those cited designs and traces how their formulations connect within the Shannon framework.

Illustrative mapping from cited studies:

Step 1: Wi-Fi sensing variables as used in cited studies

- A : Discrete activity class random variable, $A \in \{a_1, a_2, \dots, a_M\}$
- S : CSI signal observations, $S \in \mathbb{R}^{N \times T \times F}$ (N antennas, T time, F frequency)

Step 2: Model the Wi-Fi sensing ”channel” The Wi-Fi sensing process represents a communication channel:

$$A \rightarrow [\text{Human electromagnetic scattering}] \rightarrow \text{CSI observation } S$$

Step 3: Derive mutual information bounds Applying Equation 10:

$$\begin{aligned} I(A; S) &= H(A) - H(A|S) \\ &= H(A) - \sum_a P(a) H(S|A = a) \\ &\leq \log_2(M) - H_{\min}(S|A) \end{aligned}$$

where $H_{\min}(S|A)$ represents the minimum conditional entropy, determined by:

- Electromagnetic scattering properties: Human tissue dielectric constant $\epsilon_r \approx 50 - 80$ at Wi-Fi frequencies (2.4-5 GHz), where ϵ_r is the relative permittivity of human tissue compared to free space [11]
- Multipath fading: Environmental reflections causing signal variations, characterized by path loss exponent α and reflection coefficients Γ_i for different material surfaces in the environment

- Device noise: Thermal noise power $N_0 = k_B T B$ (where k_B is Boltzmann constant, T is temperature in Kelvin, B is bandwidth) and phase noise variance σ_ϕ^2 from Wi-Fi receiver oscillators

Quantitative parameter specifications:

- M : Activity class cardinality with specific ranges based on application scope:
 - Basic gesture recognition: $M = 5 - 8$ (hand gestures: swipe left/right, push/pull, rotate)
 - Full-body activity recognition: $M = 10 - 15$ (walking, sitting, standing, lying, exercising)
 - Fine-grained behavior analysis: $M = 15 - 25$ (detailed postures and micro-movements)
 - Information capacity impact: $\log_2(M)$ ranges from 2.3 bits ($M=5$) to 4.6 bits ($M=25$)
- Multipath fading quantification:
 - Path loss exponent: $\alpha = 2.0$ (free space), $\alpha = 2.5 - 3.0$ (indoor line-of-sight), $\alpha = 3.5 - 5.0$ (indoor non-line-of-sight)
 - Number of significant paths: $L = 3 - 10$ dominant multipath components in typical indoor environments
 - Reflection coefficients: $|\Gamma_{wall}| \approx 0.3 - 0.7$ (drywall), $|\Gamma_{concrete}| \approx 0.8 - 0.9$ (concrete), $|\Gamma_{metal}| \approx 0.9 - 1.0$ (metal surfaces)
 - Coherence bandwidth: $B_c = \frac{1}{5\tau_{rms}}$ where $\tau_{rms} = 10 - 100$ ns (typical indoor RMS delay spread)
- Device noise quantification:
 - Thermal noise floor: $N_0 = -174$ dBm/Hz at room temperature (290K), total noise $N_{total} = N_0 + 10 \log_{10}(B)$
 - Receiver noise figure: $NF = 6 - 12$ dB for commodity Wi-Fi receivers, $NF = 3 - 6$ dB for high-quality research equipment
 - Phase noise variance: $\sigma_\phi^2 = 10^{-6}$ to 10^{-3} rad² depending on oscillator quality and frequency offset
 - ADC quantization: $b = 8 - 16$ bits resolution, quantization noise power $\sigma_q^2 = \frac{\Delta^2}{12}$ where $\Delta = \frac{V_{range}}{2^b}$
 - Signal-to-noise ratio range: SNR = 10-40 dB in typical Wi-Fi sensing scenarios

Parameter source validation and literature evidence:

- Human tissue dielectric constants ($\epsilon_r = 50 - 80$): [11]
 - Primary source: Gabriel et al. (1996) [11] comprehensive literature survey and measurements of biological tissue dielectric properties
 - Frequency range: Covers 10 Hz to 100 GHz, directly applicable to Wi-Fi bands (2.4-5 GHz)
 - Reported values: Muscle tissue $\epsilon_r \approx 55 - 65$ at 2.4 GHz, with frequency-dependent variation
 - Note: Due to unavailability of Peyman et al. (2007) validation study, parameter uncertainty should be considered in system design

- Path loss exponent ($\alpha = 2.0 - 5.0$):
 - Theoretical foundation: Rappaport (2002) [12] wireless communication principles
 - Indoor measurements: Hashemi (2002) [13] statistical characterization of indoor radio propagation channel
 - Environmental dependence: $\alpha = 2.0$ (free space), $\alpha = 2.5 - 3.0$ (indoor LOS), $\alpha = 3.5 - 5.0$ (indoor NLOS)
 - Validation scope: Measurements across various building types and frequency bands including ISM bands
- Wi-Fi receiver noise characteristics:
 - Thermal noise foundation: Johnson-Nyquist theorem: $N_0 = k_B T$ (fundamental physical law)
 - Commercial Wi-Fi devices: Halperin et al. (2011) [14] Intel 5300 NIC characterization
 - Measured noise figure: Approximately 6-12 dB for commodity 802.11n devices based on experimental observations
 - Phase noise: Device-specific measurements required for precise characterization
- Multipath channel parameters:
 - Statistical model: Saleh & Valenzuela (1987) [15] indoor multipath propagation model
 - Delay spread measurements: RMS delay spread $\tau_{rms} = 10 - 100$ ns for typical indoor environments [15]
 - Path count: Number of significant multipath components typically 3-10 based on cluster-based channel model [15]
 - Coherence bandwidth: $B_c \approx 1/(5\tau_{rms})$ relationship from channel modeling theory

Parameter validation methodology and limitations:

1. Literature-based approach: Parameters derived from established wireless communication and biomedical literature
2. Cross-validation: Multiple independent sources confirm parameter ranges within acceptable uncertainty bounds
3. Application-specific considerations: Wi-Fi sensing may require environment-specific calibration for optimal performance
4. Uncertainty quantification: Parameter variations (± 10 -20%) should be considered in robust system design
5. Experimental validation recommended: Site-specific measurements advised for critical applications

Step 4: Wi-Fi sensing capacity as discussed in reviewed studies Reviewed studies adapt Shannon's capacity definition (Eq. 11 [8]) to bound how much activity information a CSI channel can carry:

$$C_{\text{Wi-Fi}} = \max_{p(A)} I(A; S) = \log_2(M) - H_{EM}(S|A)$$

where $H_{EM}(S|A)$ represents the intrinsic uncertainty caused by electromagnetic field scattering. Variable measurement and estimation methods:

- M : Number of activity classes, determined by application requirements (typically 5-15 activities like walking, sitting, gesturing)
- $S \in \mathbb{C}^{N \times T \times F}$: CSI measurements directly obtained from Wi-Fi devices (Intel 5300 NIC, Atheros AR9344, etc.)
 - $N = 1 - 8$: Number of antenna pairs available in Wi-Fi device hardware
 - $T = 100 - 1000$: Temporal samples in sliding window, sampling rate typically 1000 Hz for human activity recognition
 - $F = 30, 56, 114$: Number of subcarriers in 802.11n/ac/ax OFDM systems respectively
- $H_{EM}(S|A)$: Electromagnetic scattering entropy estimated through multiple approaches:
 - Empirical estimation: Calculate conditional entropy from training data using kernel density estimation or histogram methods
 - Tissue dielectric constants: Literature values $\varepsilon_r \approx 50 - 80$ at 2.4-5 GHz for human tissue, varies with hydration and frequency [11]
 - Multipath modeling: Ray-tracing simulations or measured channel impulse responses in specific environments
 - Noise characterization: Device-specific noise floor measurements, typically -90 to -70 dBm depending on receiver quality
- C_{Wi-Fi} : Theoretical capacity bound used for system design optimization and performance benchmarking, measured in bits per symbol

Practical capacity estimation approach:

1. Data collection: Gather CSI measurements across different environments, user populations, and activity conditions
2. Entropy computation: Estimate $H(A)$ and $H(A|S)$ using non-parametric density estimation methods on real Wi-Fi sensing datasets
3. Bound calculation: Compute C_{Wi-Fi} as theoretical upper bound for recognition accuracy, providing design guidelines
4. System optimization: Use capacity bound to guide antenna placement, sampling rate selection, and feature extraction strategies

Typical parameter values in Wi-Fi sensing practice:

- Wi-Fi bandwidth: 20 MHz (802.11n), 80 MHz (802.11ac), 160 MHz (802.11ax)
- CSI sampling rate: 100-1000 Hz for human activity recognition applications
- Receiver noise floor: -90 dBm (high-quality research-grade receivers), -70 dBm (commodity Wi-Fi devices)
- Human tissue dielectric: $\varepsilon_r = 65 \pm 15$ at 2.4 GHz, $\varepsilon_r = 55 \pm 10$ at 5 GHz (frequency-dependent variation) [11]

Physical Interpretation: Wi-Fi sensing capacity is fundamentally limited by the physical properties of human body electromagnetic scattering, providing a theoretical upper bound on recognition performance.

2.3 Cross-Domain Adaptation with Physical Invariance

The mutual-information framework above characterises the maximum recoverable activity information in a single domain; the cross-domain adaptation framework below addresses what happens when the source and target domains differ in their electromagnetic parameters.

2.3.1 Classical Domain Adaptation Theory

Ben-David et al. [16] formalized domain adaptation theory, providing the following bound that Wi-Fi sensing studies cite to motivate residual-gated alignment:

PAC-Bayes bound from Ben-David et al. [16]:

$$\begin{aligned} \varepsilon_T(h) &\leq \varepsilon_S(h) + \frac{1}{2}d_{\mathcal{H}\Delta\mathcal{H}}(D_S, D_T) \\ &\quad + 4\sqrt{\frac{2d \log(2m') + \log(2/\delta)}{m'}} + \lambda \end{aligned}$$

$\mathcal{H}$ -divergence from Ben-David et al. [16]:

$$d_{\mathcal{H}}(D, D') = 2 \sup_{h \in \mathcal{H}} |Pr_D[I(h)] - Pr_{D'}[I(h)]|$$

Ideal joint hypothesis from Ben-David et al. [16]:

$$\begin{aligned} h^* &= \arg \min_{h \in \mathcal{H}} \varepsilon_S(h) + \varepsilon_T(h) \\ \lambda &= \varepsilon_S(h^*) + \varepsilon_T(h^*) \end{aligned}$$

2.3.2 How Reviewed Studies Apply Physical Invariance to Domain Adaptation

Note: The steps below illustrate the reasoning used by AirFi [17] and WiSR [18] to motivate residual-gated alignment. The framing is the survey authors' synthesis of those designs; the claim that Maxwell forms remain invariant across arbitrary deployment conditions is an assumption adopted by those studies, not a universally validated result.

Step 1: Limitation of purely statistical domain adaptation (as identified in cited studies) Purely statistical methods minimize distributional distance:

$$\mathcal{L}_{\text{statistical}} = \|\mu_{\text{source}} - \mu_{\text{target}}\|_{\mathcal{H}}^2$$

Step 2: Physical reasoning adopted in reviewed studies AirFi [17] and Olivares et al. [6] argue that Maxwell equations apply in each Wi-Fi environment:

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H} \quad (\text{per domain } d, \text{ under standard assumptions})$$

Step 3: Electromagnetic signature model used in those studies Human motion is modelled as generating field perturbations parameterized by environment-specific material properties:

$$\Phi_{EM}^{(d)}(\mathbf{r}, t) = F_{\text{Maxwell}}[\text{human_motion}(t), \varepsilon_r^{(d)}, \mu_r^{(d)}]$$

Survey interpretation: AirFi and WiSR reason that while material parameters $\varepsilon_r^{(d)}, \mu_r^{(d)}$ vary across domains, the Maxwell equation forms that constrain the residual are the same. This motivates using residual magnitude as a domain-shift indicator. Whether human motion signatures are sufficiently universal across populations and environments to support this assumption broadly is an open empirical question.

(Steps 4–5 below are the survey authors’ analytical elaboration of how the Ben-David et al. bound and the physical invariance argument are combined in AirFi-style formulations; they synthesize the design logic described in Wang et al. [17] and are not drawn from a separately validated study.)

Step 4: Define physical invariance loss

$$\mathcal{L}_{physics_invariant} = \sum_d \|\Phi_{EM}^{(d)} - \Phi_{EM}^{universal}\|^2$$

where $\Phi_{EM}^{universal}$ represents the universal electromagnetic signature of human activities.

Step 5: Combined domain adaptation objective Extending Ben-David’s framework with physical constraints:

$$\mathcal{L}_{domain} = \mathcal{L}_{source} + \lambda_{adapt}\mathcal{L}_{adaptation} + \lambda_{physics}\mathcal{L}_{invariant}$$

Variable acquisition and domain adaptation implementation:

- $\mathcal{L}_{source}$: Source domain supervised loss, computed from labeled CSI data in training environment (e.g., Laboratory A with controlled conditions)
- $\mathcal{L}_{adaptation}$: Statistical alignment loss, typically MMD or adversarial loss between source and target feature distributions
- $\mathcal{L}_{invariant}$: Physical invariance loss enforcing electromagnetic field consistency across domains
- $\lambda_{adapt} \in [0.01, 1.0]$: Statistical adaptation weight, tuned through validation on target domain or cross-validation
- $\lambda_{physics} \in [0.001, 0.1]$: Physical constraint weight, typically smaller than adaptation weight to avoid numerical instability

Domain-specific parameter estimation:

- D_S, D_T : Source and target domain data distributions
 - Source domain: Controlled laboratory environment with known geometry, materials, and user demographics
 - Target domain: New deployment environment (different room, building, user population)
- $\varepsilon_r^{(d)}, \mu_r^{(d)}$: Domain-specific material properties
 - Estimation methods: Building blueprint analysis, electromagnetic site survey, or learned parameters during adaptation
 - Typical values: Office (concrete/steel): $\varepsilon_r \approx 6$, Home (wood/drywall): $\varepsilon_r \approx 3$
- $\Phi_{EM}^{universal}$: Universal electromagnetic activity signature learned across multiple source domains during training

Practical domain adaptation workflow:

1. Source training: Train Wi-Fi sensing model on labeled data from known environment with supervised loss $\mathcal{L}_{source}$
2. Target adaptation: Collect unlabeled CSI data from new deployment environment
3. Distribution alignment: Minimize statistical discrepancy between source and target feature distributions using $\mathcal{L}_{adaptation}$
4. Physical constraint enforcement: Apply electromagnetic invariance loss $\mathcal{L}_{invariant}$ to ensure Maxwell equation compliance
5. Iterative refinement: Alternate between statistical alignment and physical constraint optimization until convergence

Implementation challenges and solutions:

- Material parameter uncertainty: Use robust optimization or parameter distributions instead of fixed values
- Limited target data: Leverage few-shot learning techniques with physical priors
- Computational complexity: Use approximated electromagnetic field models for real-time adaptation
- Multi-domain generalization: Train on diverse source environments to learn more generalizable physical invariants

Interpretation note: Maxwell equations are often used as an invariance prior in cited studies; practical robustness still depends on device quality, calibration, and environment-specific validation.

3 Case-by-Case Formulations from Representative Studies

3.1 Case 1: Cross-Domain Physical Invariance Formulation

3.1.1 Original AirFi Implementation [17]

Wang et al. [17] implemented domain generalization using separate loss components:

Adversarial loss from Wang et al. [17]:

$$\mathcal{L}_{ad} = E_{h \sim p(h)}[\log D(h)] + E_{x \sim p(x)}[\log(1 - D(Q(x)))] \quad (12)$$

MMD distribution loss from Wang et al. [17]:

$$\mathcal{L}_{MMD}(Z_1, \dots, Z_n) = \frac{1}{N^2} \sum_{1 \leq i, j \leq N} MMD(Z_i, Z_j)$$

Classification loss from Wang et al. [17]:

$$\mathcal{L}_{ce} = -E_{z, y} \log[p(y|z)] \quad (13)$$

3.1.2 Formulation Notes

Step 1: Physical reasoning for cross-domain residual monitoring (as adopted in reviewed studies) Studies such as Olivares et al. [6] apply frequency-domain Maxwell equations as constraints, reasoning that their form is consistent across environments:

$$\begin{aligned}\nabla \times \mathbf{E}(\mathbf{r}, \omega) &= -j\omega\mu(\mathbf{r})\mathbf{H}(\mathbf{r}, \omega) \\ \nabla \times \mathbf{H}(\mathbf{r}, \omega) &= j\omega\varepsilon(\mathbf{r})\mathbf{E}(\mathbf{r}, \omega) + \mathbf{J}(\mathbf{r}, \omega)\end{aligned}$$

Step 2: Model human activity current density Human motion generates specific current density patterns [11]:

$$\mathbf{J}_{human}(\mathbf{r}, \omega, t) = \sigma_{tissue}(\omega) \cdot \mathbf{v}_{body}(\mathbf{r}, t) \cdot \mathbf{E}_{induced}(\mathbf{r}, \omega)$$

where $\mathbf{v}_{body}(\mathbf{r}, t)$ represents the velocity field of body parts.

Step 3: Prove invariance across domains For two different environments d_1, d_2 with the same human activity:

$$\mathbf{J}_{human}^{(d_1)}(\mathbf{r}, \omega, t) \approx k \cdot \mathbf{J}_{human}^{(d_2)}(\mathbf{r}, \omega, t)$$

where k is an environment-dependent scaling factor.

This implies CSI phase variation patterns maintain similarity [6]:

$$\Delta\phi_{CSI}^{(d_1)}(t) \approx \Delta\phi_{CSI}^{(d_2)}(t)$$

Step 4: Theoretical extension of AirFi (*The following step synthesizes the physical-invariance argument implicit in the AirFi formulation [17]; it is an analytical elaboration for this survey and is not drawn from a separate validated study.*) Building upon the original AirFi losses (Equations 12-13):

$$\begin{aligned}\mathcal{L}_{AirFi_theory} &= \mathcal{L}_{ad} + \mathcal{L}_{MMD} + \mathcal{L}_{ce} \\ &\quad + \lambda_{physics}\mathcal{L}_{EM_invariant}\end{aligned}$$

where the electromagnetic invariance loss is:

$$\mathcal{L}_{EM_invariant} = \sum_d ||\mathbf{J}_{human}^{(d)} - \mathbf{J}_{human}^{reference}||_2^2$$

3.2 Case 2: Compression and Recognition Trade-off Formulation

3.2.1 Original EfficientFi Implementation [9]

Yang et al. [9] implemented a vector-quantized variational autoencoder (VQ-VAE) with joint learning:

Reconstruction loss from Yang et al. [9]:

$$\mathcal{L}_r = ||x - D(E_c(x) + sg[E_d(x) - E_c(x)])||_2^2$$

Vector quantization loss from Yang et al. [9]:

$$\mathcal{L}_c = ||sg[E_c(x)] - E_d(x)||_2^2$$

Encoder-classifier joint loss from Yang et al. [9]:

$$\mathcal{L}_e = \lambda \|E_c(x) - \text{sg}[E_d(x)]\|_2^2 + \mathcal{L}_y(x, y)$$

Cross-entropy loss from Yang et al. [9]:

$$\mathcal{L}_y(x, y) = -E_{(x,y)} \sum_t I[y = t] \times \log \sigma(G(\hat{E}_c(x)))$$

Total EfficientFi loss from Yang et al. [9]:

$$\mathcal{L}_{EfficientFi} = \mathcal{L}_r + \mathcal{L}_c + \mathcal{L}_e$$

3.2.2 Formulation Notes

Step 1: Limitation identified in compression literature Prior studies observe that naive high-ratio compression tends to remove activity-discriminative information alongside noise [9, 19]:

Step 2: Derive an information-theoretic relationship [8] For original signal S , compressed signal S_c , and activity labels A [8]:

$$I(A; S) = I(A; S_c) + I(A; S|S_c)$$

Step 3: Model signal components

$$S = S_{activity} + S_{noise} + S_{user_specific} + S_{environment}$$

Step 4: Compression objective preserving activity-relevant information

$$S_c \approx S_{activity}$$

(Preserve activity information)

$$S_{lost} \approx S_{noise} + S_{user_specific} + S_{environment}$$

(Discard interference)

Step 5: Task-relevant representation efficiency By the data-processing inequality [8], $I(A; S_c) \leq I(A; S)$: compression cannot create new information about activity A . The benefit of compression is therefore not higher total mutual information but a reduction in nuisance variation. Under the idealized separation in Steps 1–4, the compressed signal $S_c \approx S_{activity}$ removes the components (S_{noise} , $S_{user_specific}$, $S_{environment}$) that inflate the dimensionality of S without contributing information about A , so a classifier operating on S_c faces lower effective uncertainty:

$$H(A|S_c) \approx H(A|S_{activity}) \leq H(A|S)$$

(Nuisance removal reduces effective classifier entropy)

This holds because [8]:

$$H(A|S_c) < H(A|S)$$

(Conditional entropy decreases after compression)

Step 6: Illustrative linkage to EfficientFi The EfficientFi implementation (Equation ??) is consistent with this interpretation: - $\mathcal{L}_r$ ensures signal fidelity - $\mathcal{L}_c$ creates discrete latent space for compression - $\mathcal{L}_e$ (including $\mathcal{L}_y$) ensures quantization preserves activity-discriminative information

This suggests that compression and recognition can be cooperative, depending on representation quality and deployment conditions.

3.3 Case 3: Phase Reconstruction Formulation

Note: The equations below reproduce formulations from Chen et al. [20] and Meneghello et al. [21]. The supplementary steps trace how those cited designs connect to phase information theory, following the formulations in the cited sources.

3.3.1 Original WiPhase Implementation [20]

Chen et al. [20] implemented phase-based Wi-Fi sensing:

CSI channel model from Chen et al. [20]:

$$R(f, t) = C(f, t) \times T(f, t) + N(f, t)$$

Phase measurement error model from Chen et al. [20]:

$$\angle c_{s,m}^{nt,nr} = \angle c_{s,t}^{nt,nr} + (n_p + n_s)S_s + n_c + P_{dli} + E$$

Phase ratio calculation from Chen et al. [20]:

$$pr_s^{nt,nr,nr+1} = \frac{e^{-j\angle c_{s,m}^{nt,nr+1}}}{e^{-j\angle c_{s,m}^{nt,nr}}}$$

DTW-based correlation from Chen et al. [20]:

$$DTW(D_i, D_j) = \min_P \sum_{l=1}^L ||D_i(a_l) - D_j(b_l)|| \quad (14)$$

$$c_{ij}(a_l, b_l) = ||D_i(a_l) - D_j(b_l)|| + \min[c_{ij}(a_{l-1}, b_{l-1}), c_{ij}(a_{l-1}, b_l), c_{ij}(a_l, b_{l-1})] \quad (15)$$

3.3.2 Formulation Notes

Step 1: Motivation for phase-preserving methods (as reported in reviewed studies) WiPhase [20] and SHARP [21] motivate phase-preserving design by noting that amplitude-only CSI discards the imaginary component:

$$H_{traditional} = |H(f, t)| \quad (\text{Amplitude only})$$

Complete CSI information:

$$H_{complete} = |H(f, t)|e^{j\angle H(f, t)} \quad (\text{Amplitude} + \text{Phase})$$

Step 2: Information-theoretic motivation adopted in reviewed studies WiPhase and SHARP argue that phase retains additional activity-discriminative information compared to amplitude alone, citing [8]:

$$I(A; \angle H) \geq I(A; |H|) \quad (\text{assumed in those studies; conditions not universally verified})$$

Step 3: Phase ratio formulation from WiPhase [20] From Equation ??, the phase ratio eliminates common noise:

$$\begin{aligned} pr_s^{nt,nr,nr+1} &= \frac{e^{-j(\angle c_{s,t}^{nt,nr+1} + noise_{nr+1})}}{e^{-j(\angle c_{s,t}^{nt,nr} + noise_{nr})}} \\ &\approx \frac{e^{-j\angle c_{s,t}^{nt,nr+1}}}{e^{-j\angle c_{s,t}^{nt,nr}}} \quad (\text{When noise is correlated}) \end{aligned}$$

Step 4: Graph-theoretic reading of subcarrier correlations (survey interpretation) The DTW-based correlation (Equations 14–15) captures non-linear temporal relationships between subcarriers. One interpretive reading, used here as an organising lens, is to treat subcarriers as nodes and DTW distances as edges; the resulting graph structure may reflect spatial-frequency propagation patterns, though this graph framing is the survey authors’ synthesis and is not explicitly stated in the original WiPhase paper [20].

3.4 Case 4: Sparse Geometric Modeling Formulation

Note: The equations below reproduce formulations from Meng et al. [22], with supporting compressed-sensing background from [23, 24]. The supplementary steps trace how those cited designs connect the sparse recovery objective to electromagnetic propagation structure, following the cited formulations.

3.4.1 Original WiHGR Implementation [22]

Meng et al. [22] implemented sparse recovery for dominant path selection:

CSI dynamic component separation from Meng et al. [22]:

$$H_d(f, t) = \sum_{q=1}^Q r_q \cdot e^{-j2\pi d_q(t)/\lambda}$$

Phase difference calculation from Meng et al. [22]:

$$\angle H_i^{n_R} - \angle H_i^{n_R-1} = \frac{2\pi f d \cos \theta_q}{c}$$

Steering matrix construction from Meng et al. [22]:

$$\phi_{inR}(l_q, \theta_q) = \exp \left(-j2\pi \left(\frac{(i-1)f_{ij}l_q}{c} + \frac{f(n_R-1)d \cos \theta_q}{c} \right) \right) \quad (16)$$

$$w(l_q, \theta_q) = [1, \phi_{21}(l_q, \theta_q), \dots, \phi_{INR}(l_q, \theta_q)]^T \quad (17)$$

$$W = [w(l_1, \theta_1), w(l_2, \theta_2), \dots, w(l_Q, \theta_Q)] \quad (18)$$

Sparse recovery optimization from Meng et al. [22]:

$$\min_{R_G} ||H_i - W_G R_G||_2^2 + \kappa ||R_G||_1$$

3.4.2 Formulation Notes

Step 1: Electromagnetic multipath propagation model Wi-Fi signals propagate through multiple paths [15]:

$$H(f, t) = \sum_{q=1}^{Q_{total}} \alpha_q(t) e^{-j2\pi f \tau_q(t)}$$

where Q_{total} can be hundreds of paths in complex indoor environments.

Step 2: Natural sparsity in electromagnetic domain Human activities primarily affect a small subset of dominant propagation paths:

$$Q_{dominant} \ll Q_{total} \quad (5 \text{ dominant paths from hundreds})$$

Step 3: Compressed sensing theory application The sparse recovery problem (Equation ??) follows compressed sensing theory [23, 24]:

$$\min_x \|x\|_1 \quad \text{subject to} \quad \|y - \Phi x\|_2^2 \leq \epsilon$$

where: - y : Observed CSI measurements - Φ : Measurement matrix (related to steering matrix W_G) - x : Sparse representation in geometric domain (R_G)

Step 4: Electromagnetic geometric constraints The steering matrix elements (Equation 16) incorporate wave propagation physics:

$$\begin{aligned} \phi(l_q) &= e^{-j2\pi f l_q / c} \quad (\text{Distance-dependent phase shift}) \\ \phi(\theta_q) &= e^{-j2\pi d \cos \theta_q / \lambda} \quad (\text{Angle-dependent phase shift}) \end{aligned}$$

These constraints are designed to be consistent with electromagnetic wave propagation geometry as formulated by Meng et al. [22]; whether the sparse representation fully captures all relevant propagation phenomena in practical indoor environments is treated as an open empirical question in that study.

3.5 Case 5: Feature Decoupling Formulation

Note: this case synthesizes the feature-decoupling approach discussed by WiPhase [20] and related works; the step-by-step derivation is the survey authors' analytical elaboration of principles from those studies and is not drawn from a single validated experimental paper.

3.5.1 Formulation Notes

Problem Formulation: Wi-Fi CSI contains mixed information sources:

$$\mathbf{H}_{CSI}(t) = \mathbf{F}_{gesture}(t) \oplus \mathbf{F}_{identity} \oplus \mathbf{F}_{environment} \oplus \mathbf{F}_{physics}$$

Objective: Orthogonal decomposition to isolate activity-related features.

Step 1: Information-theoretic orthogonality constraint Minimize mutual information between feature components:

$$\min I(\mathbf{F}_{gesture}; \mathbf{F}_{identity}) + I(\mathbf{F}_{gesture}; \mathbf{F}_{environment})$$

Step 2: Mathematical orthogonality enforcement

$$\mathcal{L}_{orthogonal} = ||\mathbf{F}_{gesture}^T \mathbf{F}_{identity}||_F^2 + ||\mathbf{F}_{gesture}^T \mathbf{F}_{environment}||_F^2$$

Step 3: Adversarial training for feature separation

$$\mathcal{L}_{adversarial} = -\mathcal{L}_{identity_classifier}(\mathbf{F}_{gesture}) - \mathcal{L}_{environment_classifier}(\mathbf{F}_{gesture})$$

This forces the gesture features to be non-discriminative for identity and environment classification.

3.6 Case 6: Physics-Constrained Learning Formulation

3.6.1 Composite Physics-Constrained Formulation

Building upon Supplementary Material (Maxwell-PINN formulations), this case summarizes a composite physics-constrained objective:

Composite objective pattern appearing in several reviewed formulations:

$$\mathcal{L}_{composite} = \mathcal{L}_{data} + \sum_{i=1}^4 \lambda_i \Omega_i^{physics} + \gamma \Phi_{consistency}(\mathbf{H}, \mathcal{A})$$

Four Maxwell-equation residual terms used in reviewed implementations:

$$\begin{aligned} \Omega_1^{physics} &: \text{Faraday's law} \\ \Omega_2^{physics} &: \text{Ampère-Maxwell law} \\ \Omega_3^{physics} &: \text{Gauss's law for electricity} \\ \Omega_4^{physics} &: \text{Gauss's law for magnetism} \end{aligned}$$

where $\Omega_1^{physics}$ penalises the curl of $\mathbf{E}$ (magnetic field variation), $\Omega_2^{physics}$ the curl of $\mathbf{H}$ (current and field variation), $\Omega_3^{physics}$ the divergence of $\mathbf{E}$ (charge density), and $\Omega_4^{physics}$ the divergence of $\mathbf{H}$ (no magnetic monopoles).

Maxwell Equation Constraints in Frequency Domain [6]:

$$\begin{aligned} \Omega_1^{physics} &= ||\nabla \times \mathbf{E} + j\omega\mu\mathbf{H}||^2 \\ \Omega_2^{physics} &= ||\nabla \times \mathbf{H} - j\omega\varepsilon\mathbf{E}||^2 \\ \Omega_3^{physics} &= ||\nabla \cdot (\varepsilon\mathbf{E})||^2 \\ \Omega_4^{physics} &= ||\nabla \cdot (\mu\mathbf{H})||^2 \end{aligned}$$

Consistency Function:

$$\Phi_{consistency}(\mathbf{H}, \mathcal{A}) = ||\mathbf{H}_{predicted}(\mathcal{A}) - \mathbf{H}_{observed}||^2$$

This ensures coherence between predicted CSI from activity labels and actual observations.

4 Integration Summary

4.1 Cross-Reference Map of Appendix Notes and Cases

The six appendix cases can be cross-referenced to the three technical notes as follows:

Note $\rightarrow$ case mapping:

- Maxwell-PINN note $\rightarrow$ Physics-constrained case (Case 6)
- Information-theory note $\rightarrow$ Compression case (Case 2)
- Domain-adaptation note $\rightarrow$ Physical invariance case (Case 1)
- Combined notes $\rightarrow$ Phase reconstruction (Case 3), Feature decoupling (Case 5), Sparse modeling (Case 4)

5 Appendix Algorithms

This section provides detailed pseudocode that is cited for transparency and implementation context.

5.1 Adaptive Residual Sampler (ARS)

The adaptive residual sampler implements residual-driven sampling with balanced physics/data losses for stable industrial deployment. PINNsFormer and PhysicsSolver replace MLP blocks with attention, improving stiff-operator convergence but tripling memory, as corroborated by field deployments. De et al. [25] provide error decompositions that explain why naive sampling fails. Adaptive sampling complicates scheduling on embedded hardware because residual evaluation is memory intensive: PINNsFormer-style attention over full collocation grids has been reported to multiply memory by roughly $2.8\times$ in the deployments described by Olivares-Mendez et al. [6], and industrial case studies [26] warn that training cadence must respect maintenance windows. Algorithm 2 emits the residual trajectories required to balance these factors.

Algorithm 2 ARS

Require: Collocation set $\mathcal{C}$, boundary samples $\mathcal{B}$, PDE operator $\mathcal{F}$, parameters θ

- 1: **for** each epoch **do**
 - 2: Sample collocation points proportional to residual magnitude (DPM [27])
 - 3: Update θ using Eq. (1); adapt $(\lambda_f, \lambda_b, \lambda_i)$ per adaptive loss balancing [27]
 - 4: If residuals exhibit heavy tails, activate transformer hybrids (PINNsFormer [28], Transolver [29]) on hot regions
 - 5: Log memory footprint and latency to validate industrial constraints in [6, 26]
 - 6: **end for**
-

References

- [1] C. Jiang, J. Yang, X. Li, Q. Li, X. Zhang, and J. Ren, “Risiren: Wireless sensing system attacks via metasurface,” in *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security*, 2024, pp. 3332–3345.

- [2] W. Hou and C. Wu, “Rfboost: Understanding and boosting deep wifi sensing via physical data augmentation,” *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 8, no. 2, pp. 1–26, 2024.
- [3] A. Senanian, S. Prabhu, V. Kremenetski, S. Roy, Y. Cao, J. Kline, T. Onodera, L. G. Wright, X. Wu, V. Fatemi *et al.*, “Microwave signal processing using an analog quantum reservoir computer,” *Nature Communications*, vol. 15, no. 1, p. 7490, 2024.
- [4] J. C. Wong, C. C. Ooi, A. Gupta, and Y.-S. Ong, “Learning in sinusoidal spaces with physics-informed neural networks,” *IEEE Transactions on Artificial Intelligence*, vol. 5, no. 3, pp. 985–1000, 2022.
- [5] M. Raissi, P. Perdikaris, and G. E. Karniadakis, “Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,” *Journal of Computational physics*, vol. 378, pp. 686–707, 2019.
- [6] E. Olivares, H. Ye, A. Herrero, B. A. Nia, Y. Ren, R. P. Donovan *et al.*, “Applications of information channels to physics-informed neural networks for wifi signal propagation simulation at the edge of the industrial internet of things,” *Neurocomputing*, vol. 454, pp. 405–416, 2021.
- [7] J. Shi, Y. Liu, Z. Luo, Z. Li, C. Shen, J. Zhang, G. Wang, and N. Chi, “Simplified neural network with physics-informed module in mimo visible light communication systems,” *Journal of Lightwave Technology*, vol. 42, no. 1, pp. 57–68, 2023.
- [8] T. M. Cover, *Elements of information theory*. John Wiley & Sons, 1999.
- [9] J. Yang, X. Chen, H. Zou, D. Wang, Q. Xu, and L. Xie, “Efficientfi: Toward large-scale lightweight wifi sensing via csi compression,” *IEEE Internet of Things Journal*, vol. 9, no. 15, pp. 13 086–13 095, 2022.
- [10] Z. Chen, L. Zhang, C. Jiang, Z. Cao, and W. Cui, “Wifi csi based passive human activity recognition using attention based blstm,” *IEEE Transactions on Mobile Computing*, vol. 18, no. 11, pp. 2714–2724, 2018.
- [11] C. Gabriel, S. Gabriel, and Y. Corthout, “The dielectric properties of biological tissues: I. literature survey,” *Physics in medicine & biology*, vol. 41, no. 11, p. 2231, 1996.
- [12] T. S. Rappaport, “Wireless communications—principles and practice, (the book end).” *Microwave Journal*, vol. 45, no. 12, pp. 128–129, 2002.
- [13] H. Hashemi, “The indoor radio propagation channel,” *Proceedings of the IEEE*, vol. 81, no. 7, pp. 943–968, 2002.
- [14] D. Halperin, W. Hu, A. Sheth, and D. Wetherall, “Tool release: Gathering 802.11 n traces with channel state information,” *ACM SIGCOMM computer communication review*, vol. 41, no. 1, pp. 53–53, 2011.
- [15] A. A. Saleh and R. Valenzuela, “A statistical model for indoor multipath propagation,” *IEEE Journal on selected areas in communications*, vol. 5, no. 2, pp. 128–137, 1987.
- [16] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, and J. W. Vaughan, “A theory of learning from different domains,” *Machine learning*, vol. 79, no. 1, pp. 151–175, 2010.

- [17] D. Wang, J. Yang, W. Cui, L. Xie, and S. Sun, “Airfi: Empowering wifi-based passive human gesture recognition to unseen environment via domain generalization,” *IEEE Transactions on Mobile Computing*, vol. 23, no. 2, pp. 1156–1168, 2022.
- [18] S. Liu, Z. Chen, M. Wu, C. Liu, and L. Chen, “Wisr: Wireless domain generalization based on style randomization,” *IEEE Transactions on Mobile Computing*, vol. 23, no. 5, pp. 4520–4532, 2023.
- [19] D. Cheng, L. Zhang, L. Qin, S. Wang, H. Wu, and A. Song, “Maskcae: Masked convolutional autoencoder via sensor data reconstruction for self-supervised human activity recognition,” *IEEE Journal of Biomedical and Health Informatics*, vol. 28, no. 5, pp. 2687–2698, 2024.
- [20] X. Chen, C. Li, C. Jiang, W. Meng, and W. Xiao, “Wiphase: A human activity recognition approach by fusing of reconstructed wifi csi phase features,” *IEEE Transactions on Mobile Computing*, 2024.
- [21] F. Meneghello, D. Garlisi, N. Dal Fabbro, I. Tinnirello, and M. Rossi, “Sharp: Environment and person independent activity recognition with commodity ieee 802.11 access points,” *IEEE Transactions on Mobile Computing*, vol. 22, no. 10, pp. 6160–6175, 2022.
- [22] W. Meng, X. Chen, W. Cui, and J. Guo, “Wihgr: A robust wifi-based human gesture recognition system via sparse recovery and modified attention-based bgru,” *IEEE Internet of Things Journal*, vol. 9, no. 12, pp. 10 272–10 282, 2021.
- [23] D. L. Donoho, “Compressed sensing,” *IEEE Transactions on information theory*, vol. 52, no. 4, pp. 1289–1306, 2006.
- [24] E. J. Candès and M. B. Wakin, “An introduction to compressive sampling,” *IEEE signal processing magazine*, vol. 25, no. 2, pp. 21–30, 2008.
- [25] T. De Ryck and S. Mishra, “Numerical analysis of physics-informed neural networks and related models in physics-informed machine learning,” *Acta Numerica*, vol. 33, pp. 633–713, 2024.
- [26] K. Luo, J. Zhao, Y. Wang, J. Li, J. Wen, J. Liang, H. Soekmadji, and S. Liao, “Physics-informed neural networks for pde problems: a comprehensive review,” *Artificial Intelligence Review*, vol. 58, no. 10, pp. 1–43, 2025.
- [27] J. Kim, K. Lee, D. Lee, S. Y. Jhin, and N. Park, “Dpm: A novel training method for physics-informed neural networks in extrapolation,” in *Proceedings of the AAAI conference on artificial intelligence*, vol. 35, no. 9, 2021, pp. 8146–8154.
- [28] Z. Zhao, X. Ding, and B. A. Prakash, “Pinnsformer: A transformer-based framework for physics-informed neural networks,” *arXiv preprint arXiv:2307.11833*, 2023.
- [29] H. Wu, H. Luo, H. Wang, J. Wang, and M. Long, “Transolver: A fast transformer solver for pdes on general geometries,” *arXiv preprint arXiv:2402.02366*, 2024.